

ID S2**GNSS as a sensor with UART interface (Waveshare L76K GPS HAT)****Prior knowledge**

- How to prepare, carry out, and document laboratory experiments
- Basic terms and definitions of sensor technology
- GNSS-receivers, specifically Waveshare L76K GPS HAT
(https://www.waveshare.com/wiki/L76K_GPS_HAT,
https://www.quectel.com/content/uploads/2024/02/Quectel_L76L76-L_Hardware_Design_V3.2-3.pdf)
- Function of the Universal Asynchronous Receiver Transmitter (UART)
(<https://www.electronicshub.org/basics-uart-communication/>)
- Programming the Arduino (Nano or equivalent) under the Arduino IDE
- Programming in Python (Anaconda/Spyder)
- GNSS-documentation: https://cdn.sparkfun.com/assets/a/3/2/f/a/NMEA_Reference_Manual-Rev2.1-Dec07.pdf
- Basic knowledge of Python and the libraries is an advantage (e.g. <https://py-tutorial-de.readthedocs.io/de/python-3.3/>, <https://www.python-kurs.eu/numpy.php>, ...)

Material

- Arduino Nano
- Waveshare L76K GPS HAT
- Breadboard
- USB Mini Cable
- 3 male-male Connectors
- Laptop, Arduino-IDE, Python-IDE (Anaconda), simple terminal software (z.B. HTerm)
- Arduino IDE: softwareserial (should be automatically installed with the IDE)
- Oscilloscope

Duration: 3h**Setup:**

Hardware:

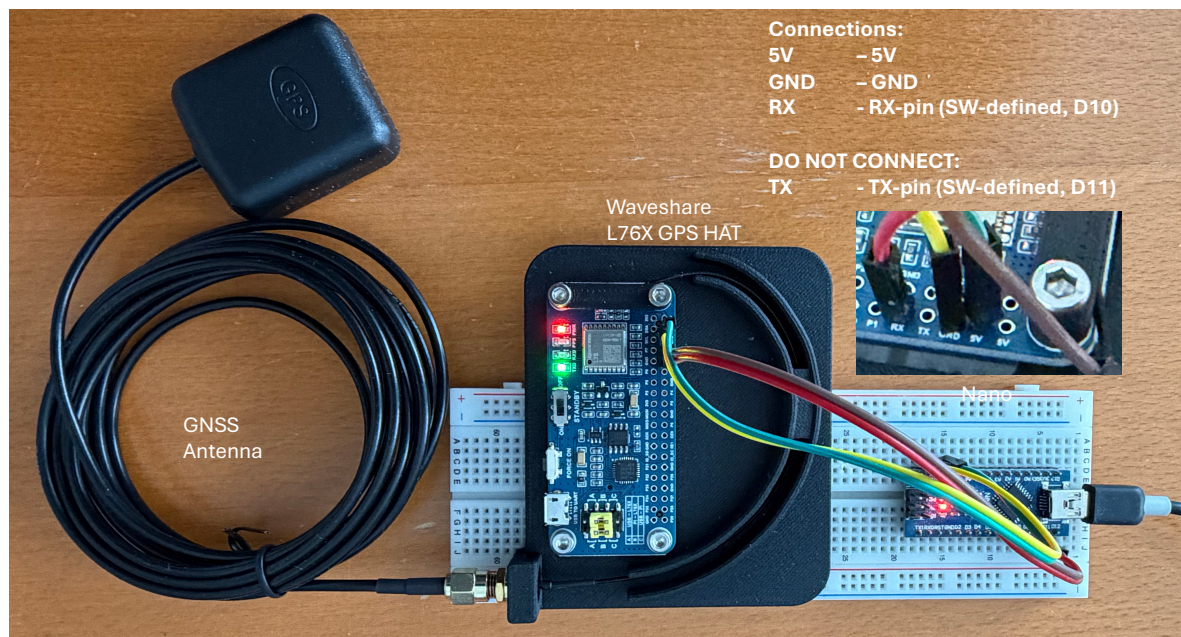


Figure 1: Electrical setup

Software:

- Arduino IDE
- Anaconda/Spyder, ...
- Prepared programs from the website (Arduino, Python):

- ArduWaveshare.ino (Arduino)
- 01_Read_USB_GNSS.py (Python, Read and Interpret GNSS-strings, collect data)

Part 1 – Setup and basic operation

- Set up the Arduino and the temperature sensor on the breadboard
- *Since the Arduino runs at 5 V for all I/O's, while the Waveshare L76K GPS HAT is operating at 3.3 V for all I/O's DO NOT CONNECT TX.* As a consequence, you can read from the UART output of the GNSS module, but you cannot write to the module.
- Three connections as indicated in Figure 1.
- Download the program "ArduWaveshare.ino" from the Teams drive, compile it and transfer it to the Arduino, check the upload process by observing the LEDs.
- Use "Tools/Serial Monitor" or "Tools/Serial Plotter" to check the sensor data; make sure that transmission parameters are set correctly (especially the baud rate) – What happens if you change the baud rate (Arduino / PC)?
- Download the program "simpleNMEA.py" and download it into your editor under Anaconda/Spyder, adjust the name of the USB-connection between computer and Arduino to your system
- Start the program there and check the output values, you may have to install the libraries "pyserial", ... in Anaconda Navigator beforehand
- The program will generate a file "GNSS.txt". Please analyze this file and answer the following questions: How does a line output by the GNSS-receiver start? Which descriptors for the following GNSS-info do you find? What is their meaning?
- Check on a map (OSM, google, ...) if the position is correct. Hint: You may have to convert latitude and longitude to Decimal Degrees!

Deliverables:

- Summary of the task in your words
- Explain the technical problem when connecting 3.3V and 5V systems through UART
- Result, confirmation of the function (photo of the set-up, screen dump)
- Create a block diagram of the structure (HW) including the electrical supply to the components and the interfaces
- Analyze both programs: How is the measurement carried out and how do the measurement results get to the laptop? How is the timestamp generated? Create a data flow diagram based on your analysis.
- Answers to
 - How does a line output by the GNSS-receiver start?
 - Which descriptors for the following GNSS-info do you find?
 - What is their meaning?
- Screendump of the map including the measured position

Part 2 – Investigation of the UART interface

- Check your oscilloscope probes for correct calibration, calibrate if required
- Connect the probes to the RX (D10) pin
- Measure voltage relative to GND as a function of time
- Keep the communication settings of both Arduino and Waveshare L76K GPS HAT at 9600 baud.
- Record at least five screen dumps with different timescales between 200ms/div and 2ms/div to see how often telegrams are transmitted and document the start of a communication in detail.
- Why does the communication still work, even though we directly connected a 3.3V to a 5 V system?
- Why would connecting the line marked TX(D11) potentially damage the Waveshare module?
- What is the time for one bit transferred?

- Annotate at least the first 10 bytes of a telegram after the Start Of Frame (setting 500us/div). Which command is transmitted?

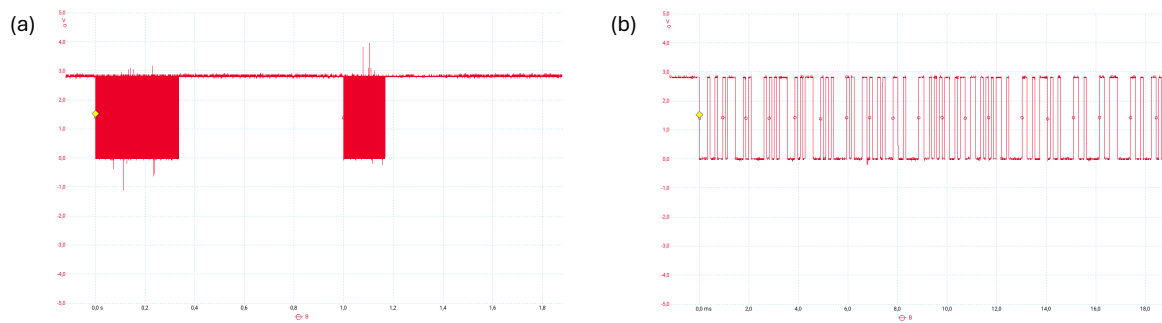


Figure 6: Examination of the UART RX/D10 with the oscilloscope (a) 200ms/div (b) 2ms/div

Deliverables:

- Summary of the task in your words
- Photo of the experimental setup
- At least five screen dumps with the above timescales
- Answers to the questions:
 - Why does the communication still work, even though we directly connected a 3.3V to a 5 V system?
 - Why would connecting the line marked TX(D11) potentially damage the Waveshare module?
 - What is the time for one bit transferred?
- Annotated screen dump at high resolution. Explanation of the values transferred.
- How often is a new position transferred to the interface?

Part 3 – Understanding GNSS-strings

- Put the antenna in a fixed position (very) close to the window; direct access to the sky is recommended; you may have to put the antenna at an angle
- Use the provided python code ("simpleNMEA.py") and run it for about 30 s.
- The program creates a text file (default "GNSS.txt")
- Analyze this text file with reference to the material given at the beginning of this document:
 - What is the structure within the text file?
 - Which lines repeat regularly?
 - What is their content?
 - Where do you find latitude and longitude of your position?

Deliverables:

- Summary of the task in your words
- Annotated/commented text file (default "GNSS.txt")
- Answers to the questions given above

Experimental part 4 – Estimating GNSS-accuracy

- Put the antenna in a fixed position (very) close to the window; direct access to the sky is recommended; you may have to put the antenna at an angle
- Use the provided python code ("NMEAReaderAccuracy.py")
- Set the parameters as required and run the program; make sure you pick a sufficiently large value for "max".
- How can you estimate the accuracy in m based on the histograms and the values given? Please perform a systematic search and choose the most suitable approach.

Deliverables:

- Task in your words
- Mean values and standard deviations for both latitude and longitude
- Histograms for latitude and longitude with correctly labeled axis'
- Your documented calculation of a position error in m